CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

TGF-β: 2 molecules stopped early

11 stopped trials across 2 molecules: Bintrafusp Alfa and Nisevokitug. 6 of the stops were the trial's own verdict and 5 followed a decision taken elsewhere. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

18.3%
11 of 60 closed trials (95% CI 10.6%–29.9%)

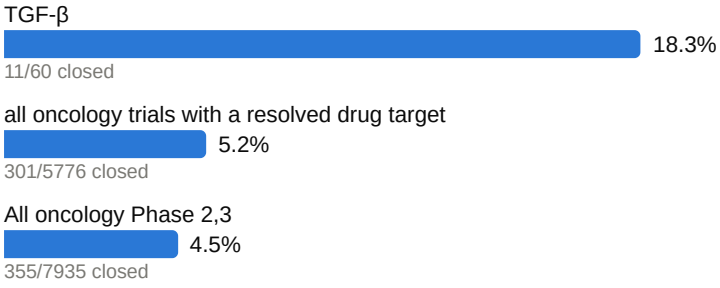
5 / 5 / 0
efficacy / safety / both (adds to 11)

50.0%
of 120 trials have closed · 60 still open or unresolved

6.4%
stopped for a biological reason within 3 years of starting (95% CI 2.9%–13.7%; 45 still at risk; 1.5% area-wide)

— **Completed and missed the primary endpoint.** None of the 33 completed trials in this segment posted a primary result on ClinicalTrials.gov that can be read, so nothing can be said here either way. 12 of the 33 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 119 distinct experimental drugs across 120 trials (protein 79, small molecule 58, antibody 26, unknown 25). Most tested: BINTRAFUSP ALFA (40), RETLIRAFUSP ALFA (19), VACTOSERTIB (13), PACLITAXEL (9), GEMCITABINE (7). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 11 stops came from 8 sponsor-asset programmes across 6 sponsors; the largest (Merck KGaA) contributed 3; removing that programme's trials from both sides leaves 15.4%. Screened alongside 85 other segments in this area, the rate stays unusual after a 10% false-discovery correction (q=0.0524, Benjamini-Yekutieli). A further 9 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 33.3%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2018–2020	7	35	20.0%	10.0%–35.9%
2021–2024	4	22	18.2%	7.3%–38.5%

Where the stops sit

Safety: 5, Efficacy / futility: 5, Benefit–risk: 1.
Sponsors with most stops: Merck KGaA (3), National Cancer Institute (NCI) (3), Novartis (2), Mayo Clinic (1), AIO-Studien-gGmbH (1), UNICANCER (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04952753	2	Novartis NVS	NISEVOKITUG, TISLELIZUMAB	2025-01	Safety	The study was terminated early due to the halt of NIS793 treatment and urgent safety measures issued in July and August 2023, respectively, as continued evaluation of Standard of Care alone will not support the trial's purpose.
NCT04390763	2	Novartis NVS	NISEVOKITUG, SPARTALIZUMAB	2024-04	Safety	The study was early terminated following the NIS793 treatment halt and urgent safety measure issued in July 2023, as the continued evaluation of Standard of Care alone will not support the original purpose of this phase 2 clinical trial.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04396535	2	Mayo Clinic	BINTRAFUSP ALFA	2023-12	Efficacy	EMD Serono recommendation based on three other studies that failed to show benefit for bintrafusp alfa
NCT03840902	2	Merck KGaA	BINTRAFUSP ALFA	2023-02	Efficacy	Based on recommendations by an external Independent data Monitoring Committee (IDMC), Sponsor decided to discontinue this clinical study due to a low likelihood of achieving superiority in the efficacy endpoints versus standard of care.
NCT04489940	2	Merck KGaA	BINTRAFUSP ALFA	2022-01	Efficacy	The study was prematurely discontinued by the sponsor due to probability of success which was too low to justify the continuation of recruitment.
NCT04327986	2	National Cancer Institute (NCI)	BINTRAFUSP ALFA, IMMUNOCYTOKINE NHS-IL12	2022-01	Benefit–risk	Study closed to accrual due to the worsening risk: benefit ratio for participants receiving bintrafusp alfa (M7824).

The 6 most recent of 11 stopped trials. Every molecule behind all 11 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 50.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.